

Coherence Field Framework (CFF)

Formal Specification

From Theoretical Foundation to Applied Discovery

Version 1.0 — May 2026

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Executive Summary

The Coherence Field Framework (CFF) is the applied successor to the Universal Field of Coherent Processes (UFCP). Where UFCP provides the theoretical physics — a conservative field framework deriving classical, quantum, and relativistic dynamics from a single Hermitian action — CFF extracts from that theory a set of **structural axioms** and converts them into **computable filters** for material and system discovery.

What CFF Is

CFF is a deterministic discovery methodology. Given a target physical property and a materials database, CFF outputs a ranked list of candidates with a fully auditable forcing chain. Every rejected candidate has a traceable reason. Every surviving candidate has a documented path through the filters.

What CFF Is Not

CFF is not a simulation tool, a machine learning model, or a curve-fitting engine. It does not predict numerical values from first principles (that is UFCP’s role). CFF identifies *architectural classes* — the structural families within which solutions must exist — by sequential constraint elimination.

Relationship to UFCP

	UFCP	CFF
Role	Theoretical foundation	Applied discovery tool
Inputs	Physical constants, action functional	Empirical databases, target property
Outputs	Equations of motion, predictions	Ranked candidates, forcing chains
Method	Variational calculus, Noether currents	Sequential constraint elimination
Free parameters	One (ρ_0)	Zero (thresholds from empirical data)
Validation	111 tests, zero falsifications	2 target domains demonstrated

Demonstrated Applications

1. **Room-temperature superconductivity (RTSC):** CFF independently derived Cu_3O_2 on $\text{LaAlO}_3(111)$ as the unique survivor of six structural filters applied to the entire known materials space. Patent filed April 2026 (US Provisional 64/037,691).
2. **Nanoscale heat transfer metamaterials:** CFF identified 12 all-phononic metamaterial geometries predicted to outperform the plasmonic-phononic approach published in *Nature* (Wang et al., 2026).
3. **Nanoparticle property prediction:** The UFCP geometric framework produced 63 quantitative predictions across 7 elements and 4 property types with 6.5% average error and zero free parameters.

4. **Phase transition detection:** The UF-Core computational kernel (L0–L4) detected phase transitions in 7 materials across 4 physical phenomena with zero parameter changes.

Part I

Theoretical Foundation (UFCP)

1 Scope

This part summarizes the UFCP theoretical framework from which CFF axioms are derived. The complete mathematical treatment is in the UFCP Comprehensive Specification v1.0 and Not-Math Framework v2.0. This section extracts only what CFF requires.

2 The Coherence Field

Definition 2.1 (Coherence Field). *Let $\psi = \sqrt{\rho} e^{i\phi}$ be the coherence field on a Lorentzian manifold $(M, g_{\mu\nu})$ with minimal gauge coupling $D_\mu = \partial_\mu + i\frac{q}{\hbar} A_\mu$.*

The field carries three physical attributes:

- $\rho(\mathbf{x}, t) \geq 0$: coherent-potential density
- $\phi(\mathbf{x}, t) \in \mathbb{R}$: phase
- $\omega = \partial_t \phi$: local tempo

Axiom 2.1 (UFCP Action).

$$S = \int d^4x \sqrt{-g} \left[\frac{i\hbar}{2} (\psi^* D_t \psi - \psi D_t \psi^*) - \psi^* \hat{\Omega}_g \psi - U(\rho) - \frac{1}{2} \int \rho W \rho' \sqrt{-g'} d^4x' \right] \quad (1)$$

where $\hat{\Omega}_g$ is a Hermitian kinetic operator, $U(\rho)$ is the local self-interaction potential (bounded below), and $W(\mathbf{x} - \mathbf{x}')$ is the symmetric interaction kernel.

Axiom 2.2 (Kernel Symmetry). $W(\mathbf{x} - \mathbf{x}') = W(\mathbf{x}' - \mathbf{x})$. Grounded in CPT invariance.

2.1 Equation of Motion

Variation of S with respect to ψ^* :

$$i\hbar D_t \psi = \hat{\Omega}_g \psi + \frac{\partial U}{\partial \rho} \psi + (W * \rho) \psi \quad (2)$$

2.2 Conserved Currents

Global $U(1)$ symmetry gives $\nabla_\mu J^\mu = 0$ with $J^0 = \rho$ and $\mathbf{J} = \rho \mathbf{v}$.

2.3 Correspondence Limits

The UFCP action reproduces exactly:

- Schrödinger equation ($U = 0$, $W = 0$, non-relativistic $\hat{\Omega}_g$)

- Newtonian mechanics (Ehrenfest limit, decoherence via W)
- Relativistic wave dynamics (Klein–Gordon / Dirac $\hat{\Omega}_g$)
- Born’s rule (from Gleason’s theorem under four structural measurement constraints)

3 The Geometric Coupling Law

Theorem 3.1 (Geometric Coupling Law).

$$Q = Q_0 (1 + \alpha^2 \lambda^2 \Gamma) \quad (3)$$

where Q_0 is the uncoupled value, $\alpha = 1/137.036$ is the fine structure constant (derived from 3D soliton stability: $\alpha = 1/N_c^2$), λ is the material-specific coupling parameter, and $\Gamma = \cos \theta$ is the geometry factor.

Ring geometry ($\theta = 0$): $\Gamma = 1$ (maximum coupling). Sphere geometry ($\theta = \pi/2$): $\Gamma = 0$ (no coupling correction). Zero free parameters.

Validation: 5 independent physical systems, 24+ materials, all consistent. See UFCP Comprehensive Spec v1.0 Part III.

4 The Soliton Ontology

Particles in UFCP are solitonic density concentrations of ψ , not fundamental point objects. The soliton width matches the Compton wavelength when $\lambda \rho_0 = mc^2/2$.

Key consequence for CFF: Structural stability of a material phase is a soliton stability problem. Phase transitions correspond to soliton bifurcations. CFF’s structural filters are constraints on soliton stability conditions.

5 Connected Constants

From the UFCP action, the following physical constants are determined (not free):

Constant	Determined by	Status
α	3D soliton stability ($1/N_c^2$)	Derived
c	$G, \rho_0, \alpha, \hbar$	Derived
G	$\lambda/(4\pi\rho_0)$	Derived
ρ_{DE}	$m\rho_0/2$	Derived
ρ_0	Initial conditions	Free parameter
$\hbar$	Coherence field quantum	Unit convention

One free parameter. One unit convention. Everything else determined.

Part II

CFF Axiom Set

6 Scope

The CFF axioms are structural constraints extracted from the UFCP action and its consequences. Each axiom generates one or more computable filters for the discovery algorithm. The axioms are numbered A1–A8.

Design principle: Each CFF axiom can be independently justified on conventional physics grounds without requiring acceptance of UFCP. CFF is designed to be auditable by domain experts who may reject the theoretical foundation entirely.

7 Axiom Definitions

Axiom 7.1 (A1 — Structural Existence). *A physical system must admit stable localized excitations (solitonic states) in its effective dimensionality. Systems whose dimensionality does not support topological closure of solitonic modes cannot sustain coherent structural phases.*

Filter: *Reject systems with dimensionality that does not support stable solitons ($d < 2$ for topological solitons; $d = 1$ supported only with specific nonlinearities).*

Conventional justification: *Mermin–Wagner theorem (no continuous symmetry breaking in $d \leq 2$ without topological protection); BKT theory (topological defects stabilize 2D systems).*

Axiom 7.2 (A2 — Norm Conservation). *The total coherent-potential density $\int \rho d^d x$ is conserved under time evolution. Systems that violate norm conservation cannot maintain structural identity.*

Filter: *Reject candidate materials or frameworks that require non-unitary evolution (e.g., open quantum systems treated as fundamental rather than effective).*

Conventional justification: *Unitarity of quantum mechanics; charge conservation; continuity equation.*

Axiom 7.3 (A3 — Boundary Condition Compatibility). *The boundary conditions of a structural system must not impose competing winding structures that destructively interfere with the target phase. Substrate–film interfaces must be thermodynamically inert with respect to the structural phase.*

Filter: *Reject substrates that (i) react with the film material, (ii) impose a competing electronic or magnetic phase, or (iii) have lattice mismatch exceeding the coherence length of the target phase.*

Conventional justification: *Epitaxial growth requires lattice matching; competing phases destroy long-range order; interfacial reactions create dead layers.*

Axiom 7.4 (A4 — Deterministic Evolution). *The structural state of a system evolves deterministically from initial conditions. The perception of structure must be reproducible: the same input field must produce the same structural description.*

Filter: *Reject stochastic methods (Monte Carlo, random initialization) as structural analysis tools. Accept only deterministic, auditable algorithms.*

Conventional justification: *Reproducibility is a foundational requirement of the scientific method.*

Axiom 7.5 (A5 — Dispositional Preservation). *Carrier tuning (doping, gating, pressure) must not disorder the coherence lattice. The structural geometry that produces the target property must be preserved under the tuning method.*

Filter: *Reject chemical doping that introduces substitutional disorder. Accept electrostatic gating, strain tuning, and interface doping that modify carrier density without disrupting the lattice.*

Conventional justification: *Chemical doping in cuprates introduces pair-breaking scattering; ionic liquid gating preserves lattice order (measured by RHEED, XRD).*

Axiom 7.6 (A6 — Non-Trivial Winding). *The structural geometry must support non-trivial phase winding — topological defects that cannot be continuously deformed to trivial configurations. Geometrically frustrated lattices satisfy this axiom; non-frustrated lattices do not.*

Filter: *Reject square, cubic, BCC, and other non-frustrated geometries. Accept kagome, triangular, honeycomb, pyrochlore, and hyperkagome geometries.*

Conventional justification: *Geometric frustration suppresses Néel ordering and enables spin-liquid, superconducting, and topological phases. Kagome lattice superconductors (CsV_3Sb_5 , Cu-BHT) have weak-coupling ratios; square-lattice cuprates have strong-coupling ratios.*

Axiom 7.7 (A7 — Coherence Preservation). *The coupling regime must preserve topological winding as the conserved quantity, not pair binding energy. Strong coupling destroys topological coherence by localizing pairs.*

Filter: *Reject systems with BCS coupling ratio $2\Delta/(k_B T_c) > r_{\max}$ (where $r_{\max}$ is target-dependent; $r_{\max} = 5.0$ for RTSC, 6.0 for HTSC).*

Conventional justification: *Known weak-coupling superconductors (MgB_2 : ratio 3.5; CsV_3Sb_5 : ratio ~ 4.0) achieve higher T_c -to-gap efficiency than strong-coupling systems (YBCO: ratio 5.6–6.0; Bi-2212: ratio 5.5–7.0).*

Axiom 7.8 (A8 — Local Field Response). *The coupling amplitude between neighboring coherent sites must exceed a threshold set by the target property. Below this threshold, structural coherence cannot propagate across the lattice at the required energy scale.*

Filter: *Reject systems with superexchange J (or equivalent coupling constant) below the target-specific threshold ($J \geq 100$ meV for RTSC, $J \geq 50$ meV for HTSC, phonon $Q \times$ oscillator strength $\geq$ threshold for thermal metamaterials).*

Conventional justification: *Superexchange J determines the pairing energy scale. Published values of J for all transition-metal oxide bridges are tabulated (YBCO Cu-O-Cu: 130 meV; NiO Ni-O-Ni: 85 meV; herbertsmithite Cu-OH-Cu: 170 meV). Only Cu-based bridges exceed 100 meV.*

8 Axiom Summary

Axiom	Name	CFF Filter	UFCP Origin
A1	Structural Existence	Dimensionality must support stable solitons	Soliton stability in d dimensions
A2	Norm Conservation	Unitary evolution required	$U(1)$ Noether current
A3	Boundary Compatibility	Substrate must be inert and lattice-matched	Boundary conditions on ψ
A4	Deterministic Evolution	Reproducible structural perception	Deterministic EoM from $\delta S = 0$
A5	Dispositional Preservation	Doping must not disorder lattice	J preserved under W perturbation
A6	Non-Trivial Winding	Frustrated geometry required	Topological soliton stability
A7	Coherence Preservation	Weak coupling regime	Winding number as conserved quantity
A8	Local Field Response	Coupling amplitude above threshold	$W * \rho$ exceeds $U(\rho)$ barrier

Part III

Computational Kernel (L0–L4)

9 Scope

The UF-Core Kernel is the computational implementation of CFF’s structural perception. It takes any dimensionless ordered field and outputs a description of that field’s structural geometry. The kernel is domain-agnostic: it does not know what the field represents.

10 Layer Architecture

Layer	Name	Function
L0	Structural Field Normalization	Converts raw input to Structural Event Vector (SEV): $(F, \Delta F, \sigma, \kappa)$. Normalized field value, first derivative, structural change rate, curvature.
L1	Gate Formation & Segmentation	Segments the field into <i>gates</i> — contiguous intervals of internally consistent structural character. Gate boundary operator: $D(t) = \alpha_1 \ \Delta F\ + \alpha_2 \sigma + \alpha_3 \kappa$. New gate opens when $D(t) < \tau_D$.
L2	Interpretive Structural Field	Maps gate sequences to interpretive vectors. Classifies each gate as STABLE, VOLATILE, TRANSITIONAL, or DEGENERATE. Performs mosaic projection.
L3	Resonance Functional	Computes resonance vector: $R_k^{\text{vec}} = (w_k, \ CV_k\ /\ CV\ _{\max}, S_k, 1/(1 + C_k), 1 - U_k)$. Unified Resonance Field: $URF_k = g_k \cdot R(k)$. Contextual gating with hysteresis.
L4	Decision Dynamics	Outputs Decision Surface Field (DSF), a 7-tuple per gate: $(D_k, M_k, U_k^*, B_k, P_k, C_k, R_{\text{rev},k})$.

11 DSF 7-Tuple (L4 Output)

Definition 11.1 (Decision Surface Field). *The DSF is the kernel’s sole output. Each gate k receives a 7-tuple:*

<i>Symbol</i>	<i>Name</i>	<i>Meaning</i>
D_k	<i>Direction</i>	<i>Structural field direction (+1, 0, -1). Reversal marks a structural direction change.</i>
M_k	<i>Momentum</i>	<i>Rate of structural change. How fast the field geometry is evolving.</i>
U_k^*	<i>Uncertainty</i>	<i>Confidence in structural assessment. Peaks at ~ 0.667 at phase transitions (maximum structural ambiguity).</i>
B_k	<i>Breathing</i>	<i>Structural freedom / expansion. How much room the field has to move.</i>
P_k	<i>Pressure</i>	<i>Transition sensitivity. Spikes when structural state is under stress.</i>
C_k	<i>Complexity</i>	<i>Mosaic divergence. How many distinct structural projections exist within a gate.</i>
$R_{rev,k}$	<i>Reversal</i>	<i>How often the structural direction flips within a gate. High = unstable.</i>

12 Domain-Agnostic Property

The same L0–L4 kernel, with identical parameters, has been applied to:

- Financial time series (TFE: 11,884 symbols, 5 years)
- Superconductor R(T) curves (7 materials, 4 phenomena)
- Pharmaceutical DSC data (polymorph screening)
- Neural network weight distributions (MACE audit)
- Nanoparticle property data (cluster physics)
- IR sensor calibration data (ArcLoom robotics)

No parameter changes between domains. The kernel perceives structure, not content.

13 Universal Uncertainty Ceiling

In every material tested, the kernel uncertainty U_k^* peaked at **0.667** at the phase transition point. This value was not programmed — it emerges from the structural geometry of the kernel when the input field is maximally ambiguous. This universality across different physical phenomena is consistent with the kernel measuring a fundamental property of structural phase transitions.

Part IV

Discovery Algorithm

14 Scope

The CFF Discovery Algorithm converts the axiom set (Part II) into a deterministic, auditable material discovery tool. Given a target physical property and empirical databases, the algorithm outputs ranked candidates with forcing chains.

15 Four-Stage Pipeline

1. **Stage 1: CFF Structural Filters.** Apply axioms A1–A8 sequentially. Each axiom generates a filter that rejects candidates violating that axiom. Coarser filters first. Each rejected candidate is logged with the specific axiom and threshold that eliminated it.
2. **Stage 2: Class Definition.** The surviving filter constraints define an *architectural class* — a structural family within which solutions must exist. This is the algorithm’s primary output.
3. **Stage 3: Class Enumeration.** Given the class definition, enumerate candidate materials from chemistry databases. Cross-reference with synthesis precedents from published literature. Rank by precedent strength.
4. **Stage 4: Output.** Produce ranked candidates with forcing chains, rejected families with reasons, and the architectural class definition. All outputs are deterministic and reproducible.

16 Required Databases

Database	Contents	Source
Bridge	Metal-ligand-metal superexchange constants J	Published measurements
Substrate	Lattice parameters, (111) surface mesh, thermodynamic inertness	Crystallography databases
Composition	Demonstrated or proposed target-geometry compositions	Published synthesis papers
Target	BCS ratio thresholds, J thresholds, domain-specific figures of merit	Empirical condensed matter

All database entries are from published empirical values. No fitted, computed, or assumed values. The algorithm’s validity rests on the quality of the databases, which are independently verifiable.

Part V

Validated Applications

17 Application 1: Room-Temperature Superconductivity

17.1 Input

Target: $T_c \geq 300$ K, ambient pressure, gateable.

17.2 Forcing Chain

1. A7 (coherence preservation) $\rightarrow$ BCS ratio $\leq 5.0 \rightarrow$ eliminates all strong-coupling superconductors
2. A6 (non-trivial winding) $\rightarrow$ frustrated geometry $\rightarrow$ eliminates square, cubic lattices
3. A8 (local field response) $\rightarrow J \geq 100$ meV $\rightarrow$ eliminates all metals except copper (Cu-O-Cu: 130 meV)
4. A6 + A8 $\rightarrow$ kagome topology $\rightarrow$ demonstrated by Möller (PCCP 2018) for Cu-O on Au(111)
5. A5 (dispositional preservation) $\rightarrow$ gate doping $\rightarrow$ eliminates chemical substitution
6. A3 (boundary compatibility) $\rightarrow$ polar oxide substrate, lattice match $\rightarrow$ LaAlO₃(111) at 5.36 Å

17.3 Output

Unique survivor: Cu₃O₂ kagome monolayer on LaAlO₃(111), electrostatically gated.

Rejected families: Square cuprates (ratio > 5), hydrides (pressure), iron-based (J too low), kagome pnictides (J too low), twisted bilayer (no high- J mechanism), nickelates (strong coupling AND low J).

Status: Patent filed (US Provisional 64/037,691, April 13, 2026). Blind predictions sealed April 29, 2026: $T_c = 370 \pm 20$ K. Awaiting experimental synthesis.

18 Application 2: Nanoscale Heat Transfer Metamaterials

18.1 Input

Target: Maximum near-field radiative heat transfer enhancement at 50–1000 nm gaps.

18.2 Forcing Chain

1. Thermal frequency $\rightarrow$ mid-IR operation $\rightarrow$ surface phonon polaritons (SPhP), not plasmons $\rightarrow$ metals eliminated as primary coupling medium

2. SPhP quality ranking $\rightarrow$ SiC forced ($Q \sim 900$, strongest oscillator strength)
3. Enhancement beyond planar $\rightarrow$ mode confinement $\rightarrow$ structured metamaterial geometry
4. Loss minimization $\rightarrow$ polar dielectric resonator, not metal $\rightarrow$ phonon-phonon coupling eliminates Ohmic loss $\rightarrow$ gold suboptimal
5. Thermal stability $\rightarrow$ all-dielectric survives $>1000^\circ\text{C}$ $\rightarrow$ gold fails at $\sim 400^\circ\text{C}$

18.3 Output

Top 3 candidates (all untested for near-field radiative heat transfer):

1. SiC pillar array on SiC substrate (self-matched phonon resonance)
2. hBN nanostructures on SiC (dual-band hyperbolic coupling)
3. SiC phononic crystal slab (slow-light flat bands)

12 total candidates enumerated. All all-phononic. All fabricable with standard cleanroom processes.

Prediction: All-phononic metamaterials outperform the plasmonic-phononic hybrid reported in *Nature* (Wang et al., 2026, $4\times$ enhancement).

Status: **UNTESTED**. Published literature supports each candidate individually. No experimental head-to-head comparison exists.

19 Application 3: Nanoparticle Property Prediction

19.1 Method

The UFCP geometric framework (not CFF filters) applied to nanoparticle cluster physics. One geometric angle $\theta = \arctan(1/\varphi)$ (icosahedral frustration, where φ is the golden ratio) generates five projection operators predicting five branches of physics.

19.2 Results

Property	Predictions	Avg Error
Magnetic moments (Fe, Co, Ni, Mn)	27	3.2%
Electron affinity (Au, Ag, Cu)	23	7.1%
Seebeck coefficient (Au_{13})	1	5.8%
HOMO-LUMO gap (Au_{13})	1	0.3%
Seebeck sign (23 elements)	23	0/23 wrong
Total	63+	6.5%

Zero free parameters. Zero exclusions. 7 elements. 5 geometries. 4 property types. All from one angle.

20 Application 4: Phase Transition Detection

20.1 Method

UF-Core kernel (L0–L4) applied to digitized R(T), dielectric, and DSC data.

20.2 Results

Material	Transition	Precursor	Detected?
YBCO	$T_c = 93$ K	13 K lead	PASS
MgB ₂	$T_c = 39$ K	18 K lead	PASS
CsV ₃ Sb ₅	$T_c = 2.5$ K	—	PASS
VO ₂	$T_{\text{MIT}} = 340$ K	20 K lead	PASS
BaTiO ₃	$T_{\text{Curie}} = 393$ K	22 K lead	PASS
FeSe/STO	$T_c = 65$ K	20 K lead	PASS
Pharma DSC	5 transitions	—	5/5 PASS

Same kernel, same parameters, all domains.

Part VI

Mathematical Notation and Glossary

21 Symbols Used

Symbol	Name	Definition
ψ	Coherence field	$\psi = \sqrt{\rho} e^{i\phi}$; complex scalar field on $(M, g_{\mu\nu})$
ρ	Coherent-potential density	$\rho = \psi ^2 \geq 0$
ϕ	Phase	$\phi \in \mathbb{R}$; local phase of coherence field
ω	Local tempo	$\omega = \partial_t \phi$; rate of phase evolution
S	Action	UFCP action functional (Axiom 2.1)
$\hat{\Omega}_g$	Kinetic operator	Hermitian operator on $(M, g_{\mu\nu})$; reduces to $-\hbar^2 \nabla^2 / (2m) + V$ in non-relativistic limit
$U(\rho)$	Self-interaction	Local potential, bounded below; focusing: $U = -\lambda \rho^2 / 2$
W	Interaction kernel	Symmetric two-point density coupling; $W(\mathbf{x} - \mathbf{x}') = W(\mathbf{x}' - \mathbf{x})$
D_μ	Gauge-covariant derivative	$D_\mu = \partial_\mu + i(q/\hbar)A_\mu$
J^μ	Noether current	Conserved current from $U(1)$ symmetry: $J^0 = \rho$, $\mathbf{J} = \rho \mathbf{v}$
α	Fine structure constant	$\alpha = 1/N_c^2 \approx 1/137.036$; from 3D soliton stability
N_c	Critical soliton number	$N_c \approx 11.7$; critical L^2 norm of 3D focusing NLS ground state
Γ	Geometry factor	$\Gamma = \cos \theta$; ring ($\Gamma = 1$), sphere ($\Gamma = 0$)
λ	Material coupling	Material-specific coupling parameter (e.g., electron-phonon coupling)
Q_0	Uncoupled observable	Standard value without geometric coupling correction
J	Superexchange constant	Energy scale of magnetic coupling across metal-ligand-metal bridge (meV)
Δ	Superconducting gap	Pairing gap energy
T_c	Critical temperature	Phase transition temperature
T_{BKT}	BKT temperature	$\pi \hbar^2 n_s / (8m^* k_B)$; 2D superfluid transition
n_s	Carrier sheet density	2D carrier concentration (cm^{-2})
m^*	Effective mass	Quasiparticle effective mass
Kernel Symbols		
SEV	Structural Event Vector	L0 output: $(F, \Delta F, \sigma, \kappa)$

Symbol	Name	Definition
DSF	Decision Surface Field	L4 output: $(D_k, M_k, U_k^*, B_k, P_k, C_k, R_{\text{rev},k})$
D_k	Direction	Structural field direction (+1, 0, -1)
M_k	Momentum	Rate of structural change
U_k^*	Adjusted uncertainty	Structural confidence; peaks at 0.667 at transitions
B_k	Breathing	Structural expansion/contraction
P_k	Pressure	Transition sensitivity
C_k	Complexity	Mosaic divergence
$R_{\text{rev},k}$	Reversal rate	Structural direction flip frequency
CFF Filter Symbols		
r_{max}	BCS ratio threshold	Maximum $2\Delta/(k_B T_c)$ for target (5.0 for RTSC)
J_{min}	Exchange threshold	Minimum superexchange for target (100 meV for RTSC)
Q_{SPhP}	Phonon polariton quality	ω_0/γ for surface phonon polariton
S_{osc}	Oscillator strength	Relative phonon oscillator strength

22 Acronyms

Acronym	Expansion
CFF	Coherence Field Framework
UFCP	Universal Field of Coherent Processes
UF-Core	Universal Field Core (L0–L4 kernel)
SEV	Structural Event Vector (L0 output)
DSF	Decision Surface Field (L4 output)
SPhP	Surface Phonon Polariton
SPP	Surface Plasmon Polariton
HPhP	Hyperbolic Phonon Polariton
SES	Secure Envelope System
SCE	Structural Cryptographic Engine
RTSC	Room-Temperature Superconductivity
HTSC	High-Temperature Superconductivity
BKT	Berezinskii–Kosterlitz–Thouless
BCS	Bardeen–Cooper–Schrieffer
NLS	Nonlinear Schrödinger (equation)
CPT	Charge–Parity–Time (symmetry)
NFRHT	Near-Field Radiative Heat Transfer
MBE	Molecular Beam Epitaxy
RHEED	Reflection High-Energy Electron Diffraction
XPS	X-ray Photoelectron Spectroscopy
RIE	Reactive Ion Etching
FIB	Focused Ion Beam

Part VII

Document Control

23 Version History

Version	Description
1.0 (May 2026)	Initial formal specification. Consolidates UFCP theoretical foundation (Not-Math v2.0, UFCP Comprehensive Spec v1.0), CFF axiom set (A1–A8), UF-Core kernel architecture (L0–L4), discovery algorithm (4-stage pipeline), and four validated applications (RTSC, metamaterials, nanoparticles, phase transitions). Mathematical glossary and acronym table included.

24 Source Documents

This specification consolidates and supersedes:

1. Not-Math Framework v2.0 (April 2026) — theoretical foundation
2. UFCP Comprehensive Specification v1.0 (May 2026) — full theoretical treatment with 111 validations
3. CFF Discovery Algorithm Spec v1.0 (May 2026) — RTSC application and 4-stage pipeline
4. CFF Metamaterial Heat Transfer Analysis v1.0 (May 2026) — thermal metamaterial application
5. UF-Core Kernel Spec with SES (Feb 2026) — L0–L4 layer definitions and security envelope
6. DSF-AI Phase Transition Service Spec v1.0 (May 2026) — commercial service architecture
7. KERNEL_PHILOSOPHY.md (May 2026) — operational constraints on kernel modification

25 Intellectual Property

- The CFF axiom set (A1–A8) is a **trade secret**.
- The UF-Core kernel (L0–L4) implementation is a **trade secret**.
- The DSF 7-tuple field definitions and their structural interpretation are **trade secrets**.
- The CFF Discovery Algorithm forcing chain methodology is a **trade secret**.
- The UFCP geometric coupling law and its applications are **trade secrets**.
- US Provisional Patent 64/037,691 (filed April 13, 2026) covers the Cu_3O_2 kagome superconductor design.
- Published results (transition temperatures, precursor onsets, regime maps, candidate rankings) are factual observations and do not expose trade secrets.

disclosure is prohibited. The theoretical foundation (UFCP) and applied methodology (CFF) are proprietary. All empirical databases use published values and are independently verifiable. The kernel implementation and axiom derivations are not disclosed in any public-facing material.